

Question				Marking details	Marks Available					
					AO1	AO2	AO3	Total	Maths	Prac
4	(a)	(i)		Slow down neutrons (1) So they {cause fission of uranium / can be captured by an uranium (atom) / can be captured by an uranium (nucleus)} (1)	2			2		
		(ii)		Any 3 × (1) from: Control rods may no longer be free to move (up and down in the channels) (1) If rods are stuck out then the reactor may start to overheat (1) since they will not be able to absorb {sufficient / enough} neutrons (to reduce rate of fission) (1) leading to possible meltdown / uncontrollable chain reaction} / you can't shut the reactor down (1) OR converse	3			3		
	(b)			${}_{36}^{92}\text{Kr} + 3$ (1)		3		3	2	
	(c)	(i)		Correct construction line(s) added (1) 700 million years (1)		2		2	2	
		(ii)		Sequence of halving – expect $1 \rightarrow \frac{1}{2} \rightarrow \frac{1}{4} \rightarrow \frac{1}{8} \rightarrow \frac{1}{16} \rightarrow \frac{1}{32}$ (1) 5 half-lives (1) $5 \times 700\,000\,000(\text{ecf}) = 3\,500\,000\,000$ years (1)		3		3	3	
				Question 4 total	5	8	0	13	7	0